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**Ex 13: Minimax Algorithm for Gaming**

**CODE:**

```
def minimax(depth, node_index, is_max, values, max_depth):
  if depth == max_depth:
    return values[node_index]
  if is_max:
    return max(minimax(depth + 1, node_index * 2, False, values, max_depth),
           minimax(depth + 1, node_index * 2 + 1, False, values, max_depth))
  else:
    return min(minimax(depth + 1, node_index * 2, True, values, max_depth),
           minimax(depth + 1, node_index * 2 + 1, True, values, max_depth))
leaves = [3, 5, 2, 9, 12, 5, 23, 23]
max_depth = 3
print("Optimal utility score:", minimax(0, 0, True, leaves, max_depth))
```

**OUTPUT:**

```
============== RESTART: C:/Users/Saiko/Documents/CSA1738/EXP-13.py =============
Optimal utility score: 12
```